WORK-STUDY FROM SEPTEMBER 2026 · POWER ELECTRONICS & ELECTRICAL ENERGY SYSTEMS

Khalil LAGHA

M2 student — Electrical Energy Systems Design (CSEE) · Université Grenoble Alpes

Ranked 2nd of the M1 EEA cohort (15.43/20). DC-DC converters (buck, boost), regulation-loop tuning, electrical machines and power grids — from the lab (GIPSA-lab) to the industrial field. Work-study rotation: 2–3 weeks company / university.

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- Authorized to work in France

01 Experience

Research Intern · GIPSA-lab, Grenoble — 04 – 07/2026 · 3.5 months

- Developed a Matlab/Simulink model of a PEMFC fuel cell — **6%** dynamic / **1%** static error.
- Tuned the regulation loop of a **boost DC-DC converter** (PI controller), targeting conversion-efficiency optimization — validated in simulation then on a real test bench.

Intern · SLB (Schlumberger) — 07/2025 · 1 month

- Wrote technical syntheses in an international industrial environment; earned the **NEST & SIPP Level 2** HSE certifications.

Power Electronics & Automation Intern · EURL Lagha — 01/2024 · 15 d + 09/2024 · 1 month

- Contributed to an industrial **buck DC-DC converter** — 50 V input, adjustable 0–48 V output, **25 A**.
- Worked on **4 transformer-rectifiers** for pipelines: 3-phase 400 V → 100 V, rectification + filtering (ripple < 5%, EMI compliance) — output **100 V DC / 80 A (≈ 8 kW)**.
- Co-designed the supervision HMI (Siemens TIA Portal); produced the electrical schematics in EPLAN.

Power Grid Intern · Sonelgaz — 03/2024 · 15 d · observation

- Studied substation structure and the full generation → distribution chain (**400/220 kV → 60 kV → 30/10 kV → 400/230 V**); visited an MV/LV substation.

02 Projects

Power grid analysis · Matlab · solo · github.com/KhalilEllahLAGHA
Load flows from **3 to 118 buses** (IEEE 14/30/57/118) by Gauss-Seidel, Newton-Raphson and fast-decoupled; direct Zbus = inv(Ybus) to 1e-12; critical fault-clearing-time study in transient stability.

Winding and flux analysis program · Python / PyQt · FEMM
Three-phase winding, tooth-based MMF, reluctance-network flux density, finite-element validation.

Machine control · Matlab/Simulink
FOC of an induction motor (Park, PID, observers); cascade control of a DC motor through a three-phase thyristor bridge (outer speed loop, inner current loop).

Neural network from scratch (MLP) · Matlab · open source (MIT)
Multilayer perceptron and backpropagation hand-coded, no toolbox — debugged, tested and published: github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab.

2nd	**15.43**	**C2**
of the M1 EEA cohort (UGA)	M1 annual average /20	English

03 Education

M2 CSEE — Electrical Energy Systems Design — 2026 –
Université Grenoble Alpes · work-study

M1 EEA (EECS) — 2025 – 26
Université Grenoble Alpes · 2nd in class

Engineering cycle, Electrical Engineering — 2021 – 25
École Nationale Polytechnique d'Alger · 4 of 5 years completed; prep classes: top-ranked

Baccalauréat in Mathematics — 2021
Highest honours — 16.93/20

04 Skills

- Power electronics
- DC-DC / AC-DC conversion
- Machines & transformers
- Grids: load flow, stability
- Loop tuning (PID, LQR)
- PLC / HMI / SCADA
- Electrical CAD: EPLAN, AutoCAD
- AI: NN, GA, fuzzy logic

05 Software

Matlab/Simulink · Python · C/C++ · LTspice · Proteus · EPLAN · AutoCAD Electrical · Siemens TIA Portal · Step7 · FEMM · SolidWorks · Arduino · Git/GitHub

06 Languages & certification

French	C1
English	C2
Arabic	native

SLB **NEST & SIPP Level 2** — 07/2025 (HSE)

07 Soft skills

- Fast learner
- Rigorous
- Organized
- Problem solving
- Comfortable with tools

Portfolio: **khalilellahlagha.github.io** · Code: **github.com/KhalilEllahLAGHA** · IEEE Student Branch ENP · VIC